

ISyE 6525 HW1 - Question 1

For the spline in the question, the outside expression is

$$y(x) = \beta_1 + \beta_2 x + \sum_{k=1}^K \theta_k (x - \xi_k)_+^2, \quad x \text{ outside } (\xi_1, \xi_K),$$

where

$$(x - \xi_k)_+ = \max(x - \xi_k, 0).$$

We want $y''(x) = 0$ outside the boundary knots.

Left of the first knot

If $x < \xi_1$, then every truncated term is zero. Therefore,

$$y(x) = \beta_1 + \beta_2 x$$

and hence

$$y''(x) = 0.$$

Thus the left boundary condition is already satisfied.

Right of the last knot

If $x > \xi_K$, then $(x - \xi_k)_+ = x - \xi_k$ for every k . Differentiating twice gives

$$y'(x) = \beta_2 + 2 \sum_{k=1}^K \theta_k (x - \xi_k),$$
$$y''(x) = 2 \sum_{k=1}^K \theta_k.$$

Therefore, the right boundary condition requires

$$\sum_{k=1}^K \theta_k = 0.$$

Equivalently,

$$\theta_K = - \sum_{k=1}^{K-1} \theta_k.$$

Substitute this into the truncated-power part:

$$\begin{aligned} \sum_{k=1}^K \theta_k (x - \xi_k)_+^2 &= \sum_{k=1}^{K-1} \theta_k (x - \xi_k)_+^2 + \theta_K (x - \xi_K)_+^2 \\ &= \sum_{k=1}^{K-1} \theta_k (x - \xi_k)_+^2 - \left(\sum_{k=1}^{K-1} \theta_k \right) (x - \xi_K)_+^2 \\ &= \sum_{k=1}^{K-1} \theta_k [(x - \xi_k)_+^2 - (x - \xi_K)_+^2]. \end{aligned}$$

Define the new basis functions

$$h_k(x) = (x - \xi_k)_+^2 - (x - \xi_K)_+^2, \quad k = 1, \dots, K-1.$$

For $x > \xi_K$,

$$\begin{aligned} h_k(x) &= (x - \xi_k)^2 - (x - \xi_K)^2 \\ &= 2(\xi_K - \xi_k)x + (\xi_k^2 - \xi_K^2), \end{aligned}$$

which is linear in x . For $x < \xi_1$, $h_k(x) = 0$. Thus $h_k''(x) = 0$ on both outside regions.

To follow the piecewise form given in the question, also define

$$q(x) = \begin{cases} x^2, & x \in [\xi_1, \xi_K], \\ 0, & x \text{ outside } (\xi_1, \xi_K). \end{cases}$$

Then the spline can be written as

$$y(x) = \beta_1 + \beta_2 x + \beta_3 q(x) + \sum_{k=1}^{K-1} \theta_k h_k(x).$$

Therefore, one set of basis functions is

$$\boxed{1, \quad x, \quad q(x), \quad h_1(x), \dots, h_{K-1}(x).}$$

This basis builds the constraint into the representation, so the remaining coefficients are unrestricted.